Forecasting

June 15, 2026

1 Chocolate Sales Forecasting

By John Osorio

The goal is to build a model to predict the future chocolate sales revenue (Amount) using historical data.

Based on the public dataset *Chocolate Sales Data* https://www.kaggle.com/datasets/atharvasoundankar/chocolate-sales

1.1 1. Imports and data loading

[281]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split

from sklearn.metrics import root_mean_squared_error
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import r2_score

from sklearn.linear_model import LinearRegression
from sklearn.linear_model import Ridge
from sklearn.linear_model import Lasso
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor

from statsmodels.tsa.seasonal import seasonal_decompose

from prophet.diagnostics import cross_validation, performance_metrics
from prophet import Prophet
```

[282]:
```python
df = pd.read_csv("chocolate_sales.csv")
print(f'rows: {df.shape[0]}, cols: {df.shape[1]}')
```

```
rows: 1094, cols: 6
```

[283]:
```
df.head(5)
```

[283]:
```
       Sales Person    Country             Product       Date    Amount  \
0  Jehu Rudeforth         UK       Mint Chip Choco  04-Jan-22    $5,320
1      Van Tuxwell      India        85% Dark Bars  01-Aug-22    $7,896
2     Gigi Bohling      India  Peanut Butter Cubes  07-Jul-22    $4,501
3     Jan Morforth  Australia  Peanut Butter Cubes  27-Apr-22   $12,726
4   Jehu Rudeforth         UK  Peanut Butter Cubes  24-Feb-22   $13,685

   Boxes Shipped
0            180
1             94
2             91
3            342
4            184
```

[284]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1094 entries, 0 to 1093
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Sales Person   1094 non-null   str
 1   Country        1094 non-null   str
 2   Product        1094 non-null   str
 3   Date           1094 non-null   str
 4   Amount         1094 non-null   str
 5   Boxes Shipped  1094 non-null   int64
dtypes: int64(1), str(5)
memory usage: 105.1 KB
```

[285]:
```
df.describe().T
```

[285]:
```
                count        mean         std  min   25%    50%     75%    max
Boxes Shipped  1094.0  161.797989  121.544145  1.0  70.0  135.0  228.75  709.0
```

[286]:
```
df.isna().sum()
```

[286]:
```
Sales Person     0
Country          0
Product          0
Date             0
Amount           0
Boxes Shipped    0
dtype: int64
```

[287]:
```
print("Duplicated rows: ", df.duplicated().sum())
```

```
Duplicated rows:  0
```

Correct format for dates and Amount

[288]:
```
df["Date"] = pd.to_datetime(df["Date"], format='%d-%b-%y')
df["Amount"] = df["Amount"].replace("[$,]", "", regex=True).astype(float)

df = df.rename(columns={ "Sales Person": "Sales_person", "Boxes Shipped":␣
↪"Boxes_shipped"})

target_col = 'Amount'
cat_col = ["Sales_person","Country", "Product"]
num_col = ["Boxes_shipped"]
```

[289]:
```
df.head(5)
```

[289]:
```
     Sales_person    Country             Product       Date   Amount  \
0  Jehu Rudeforth         UK      Mint Chip Choco 2022-01-04   5320.0
1     Van Tuxwell      India       85% Dark Bars 2022-08-01   7896.0
2    Gigi Bohling      India  Peanut Butter Cubes 2022-07-07   4501.0
3    Jan Morforth  Australia  Peanut Butter Cubes 2022-04-27  12726.0
4  Jehu Rudeforth         UK  Peanut Butter Cubes 2022-02-24  13685.0

   Boxes_shipped
0            180
1             94
2             91
3            342
4            184
```

1.2 2. Exploratory Data Analysis

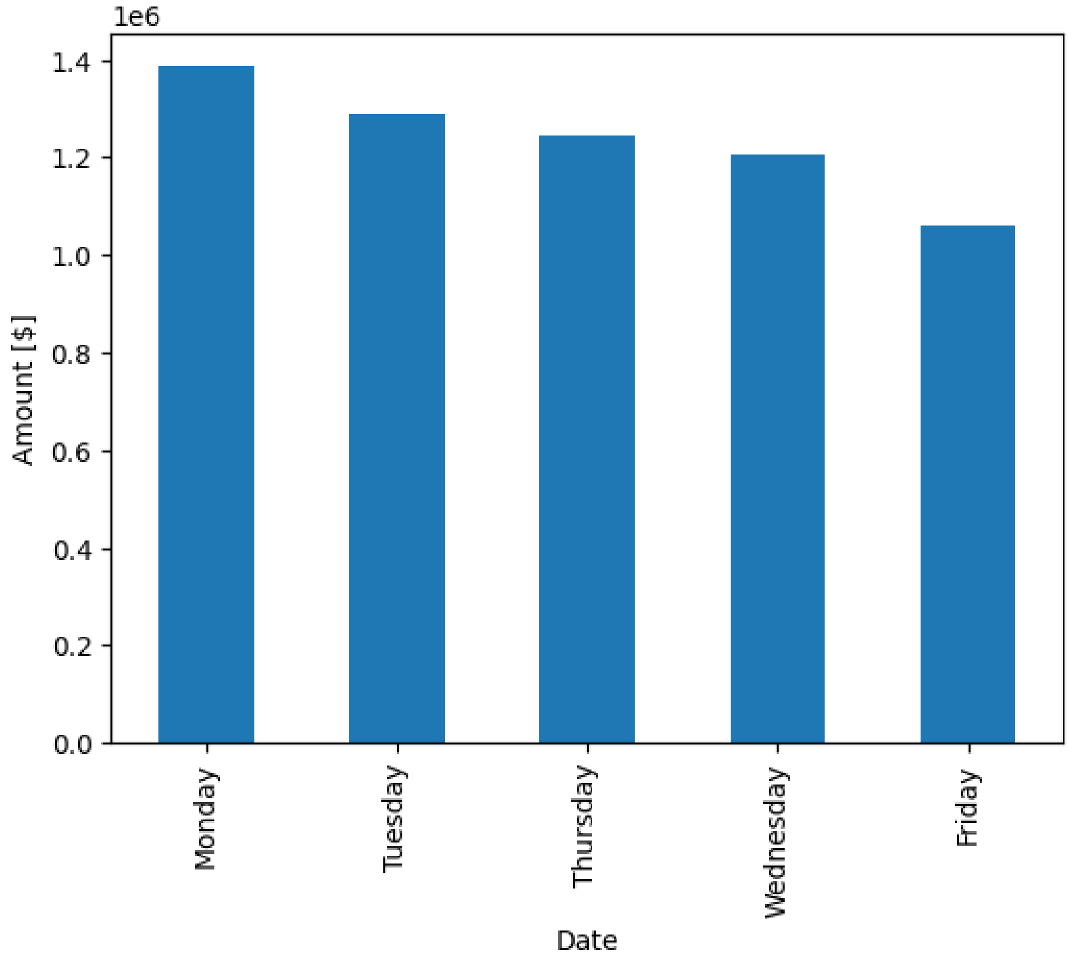
Analyzing the range of the dates and entries by day of the week

[290]:
```
print(f"Date range: [{df['Date'].min()}, {df['Date'].max()}]")
```

```
Date range: [2022-01-03 00:00:00, 2022-08-31 00:00:00]
```

[291]:
```
df.groupby(df["Date"].dt.day_name())["Amount"].sum().
↪sort_values(ascending=False).plot(kind="bar")
plt.ylabel("Amount [$]")
```

[291]:
```
Text(0, 0.5, 'Amount [$]')
```

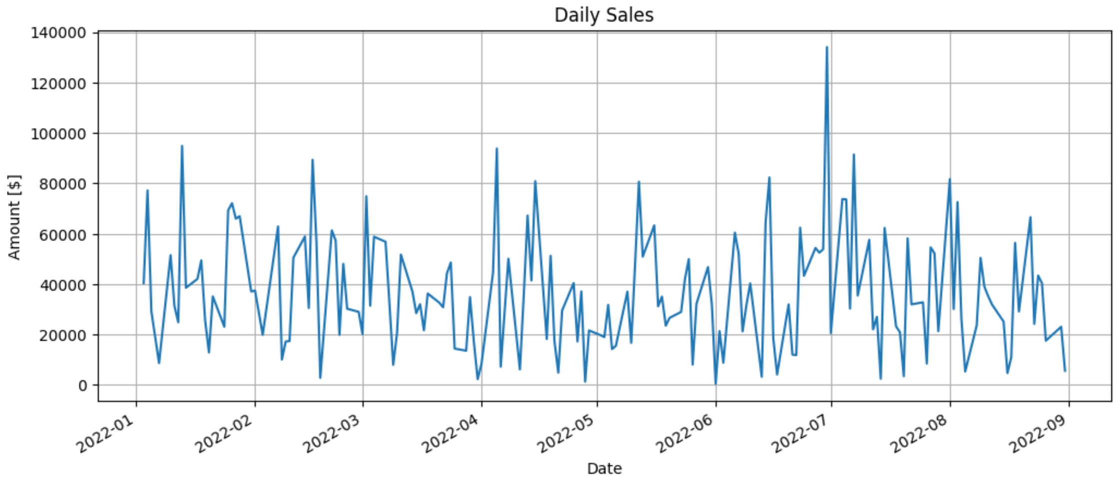


The sales data is recorded only on business days and is recorded over an 8-month period between February and August.

Target Time Series

[292]:
```
daily_sales = df.groupby("Date")["Amount"].sum()

daily_sales.plot(figsize=(12,5))
plt.title("Daily Sales")
plt.ylabel("Amount [$]")
plt.grid()
plt.show()
```

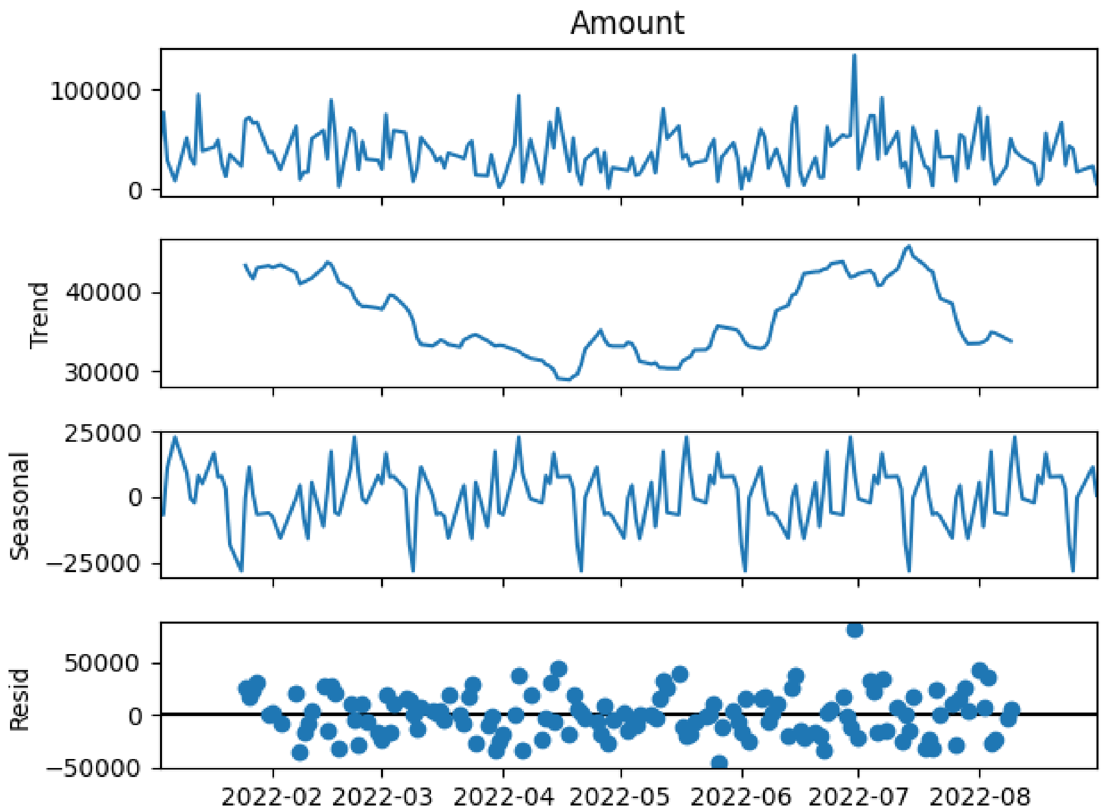


[293]:
```
daily_sales
```

[293]:
```
Date
2022-01-03    40425.0
2022-01-04    77175.0
2022-01-05    29162.0
2022-01-07     8666.0
2022-01-10    51471.0
                ...
2022-08-24    43400.0
2022-08-25    40341.0
2022-08-26    17556.0
2022-08-30    23072.0
2022-08-31     5614.0
Name: Amount, Length: 168, dtype: float64
```

[294]:
```
decompose = seasonal_decompose(daily_sales, model='additive', period=30)
decompose.plot().show()
```

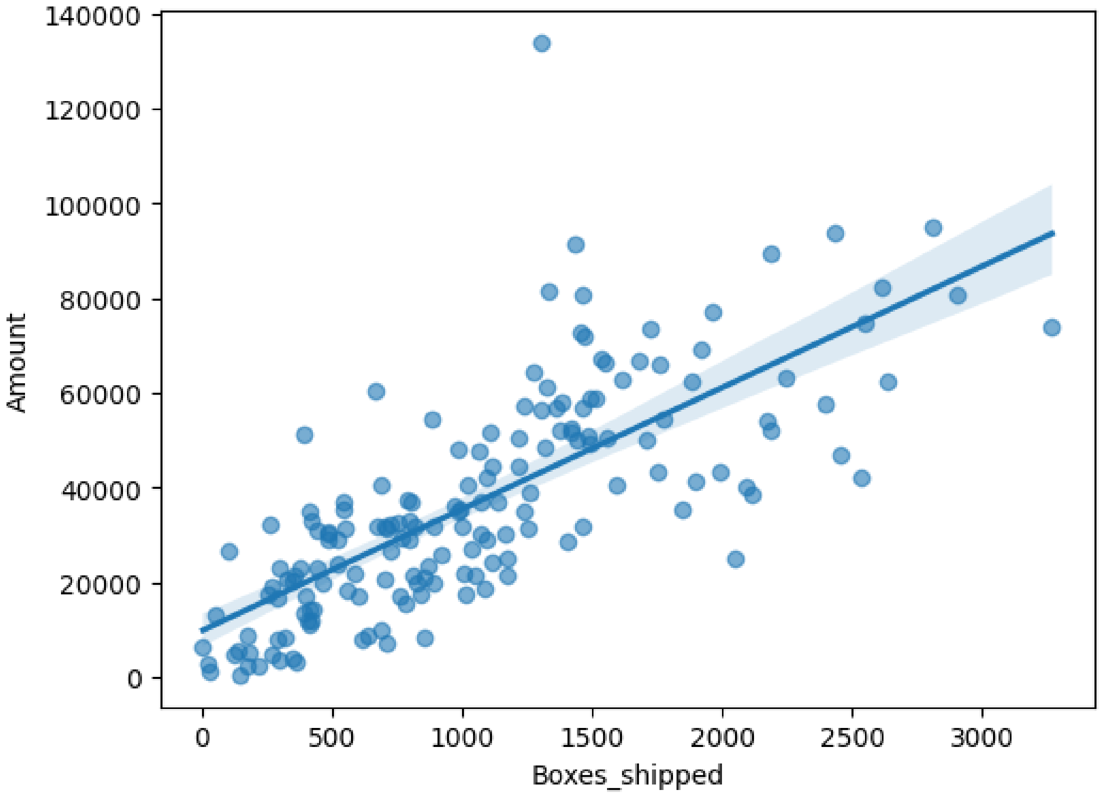
```
C:\Users\ASUS\AppData\Local\Temp\ipykernel_33920\1332662130.py:2: UserWarning:
Matplotlib is currently using module://matplotlib_inline.backend_inline, which
is a non-GUI backend, so cannot show the figure.
  decompose.plot().show()
```



Relationship between Amount and Boxes shipped

[295]:
```
df_daily = df.groupby("Date")[["Boxes_shipped", "Amount"]].sum().reset_index()
sns.regplot(data=df_daily, x="Boxes_shipped", y="Amount", scatter_kws={"alpha":
  ↪0.6})
```

[295]: `<Axes: xlabel='Boxes_shipped', ylabel='Amount'>`



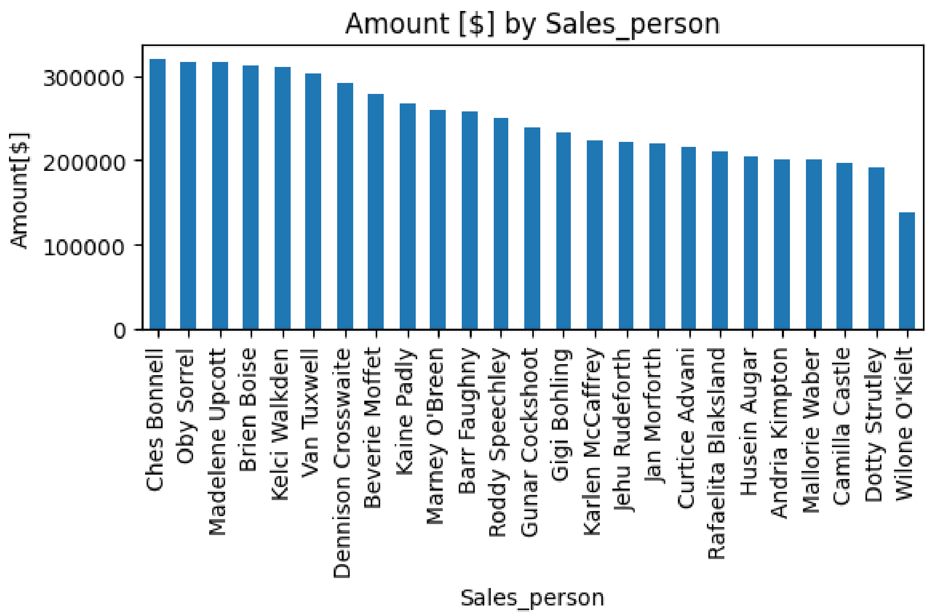
There is a positive relationship between the number of boxes shipped and the amount, as expected. A linear relationship is observed. However, the dispersion in the data suggests that the unit price is dynamic.

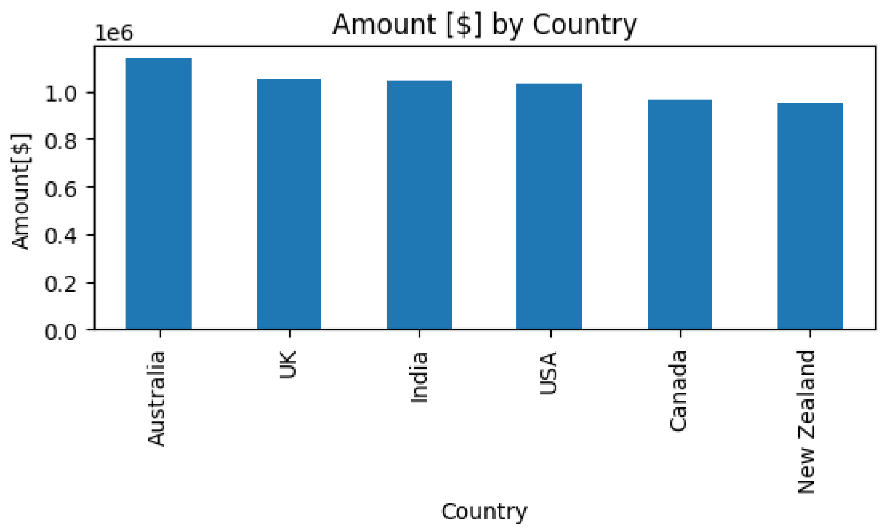
Insights of the categorical columns

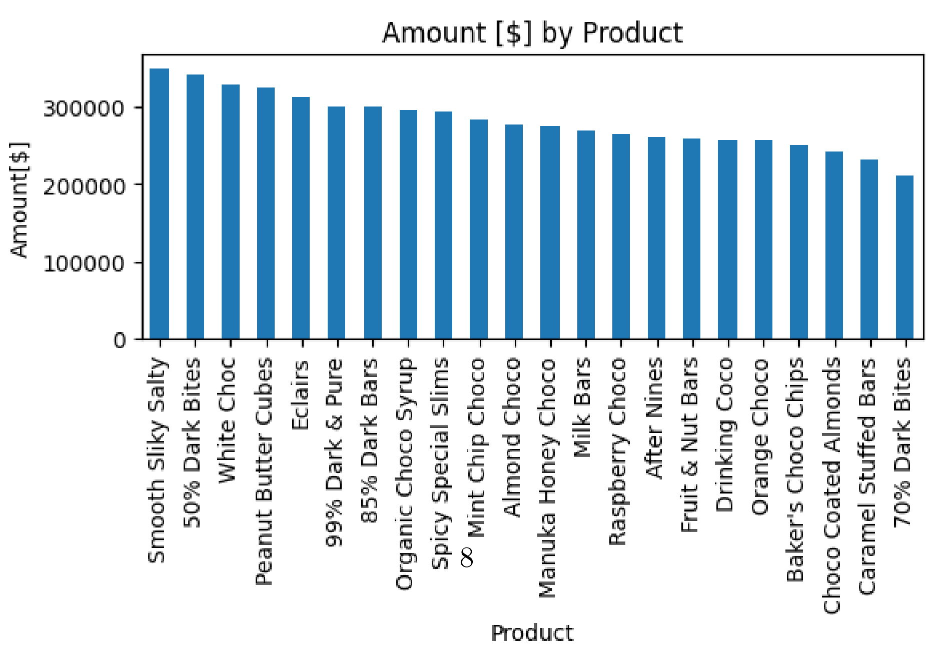
[296]:
```
fig, axes = plt.subplots(3, 1, figsize=(6, 12))

for i, col in enumerate(cat_col):
    df.groupby(col)["Amount"].sum().sort_values(ascending=False).
  ↪plot(kind="bar", ax=axes[i])
    axes[i].set_title(f"Amount [$] by {col}")
    axes[i].set_xlabel(col)
    axes[i].set_ylabel("Amount[$]")

plt.tight_layout()
plt.show()
```


Amount [$] by Sales_person
Amount[$]
0
100000
200000
300000
Ches Bonnell
Oby Sorrel
Madelene Upcott
Brien Boise
Kelci Walkden
Van Tuxwell
Dennison Crosswaite
Beverie Moffet
Kaine Padly
Marney O'Breen
Barr Faughny
Roddy Speechley
Gunar Cockshoot
Gigi Bohling
Karlen McCaffrey
Jehu Rudeforth
Jan Morforth
Curtice Advani
Rafaelita Blaksland
Husein Augar
Andria Kimpton
Mallorie Waber
Camilla Castle
Dotty Strutley
Wilone O'Kielt
Sales_person


Amount [$] by Country
1e6
Amount[$]
0.0
0.2
0.4
0.6
0.8
1.0
Australia
UK
India
USA
Canada
New Zealand
Country


Amount [$] by Product
Amount[$]
0
100000
200000
300000
Smooth Sliky Salty
50% Dark Bites
White Choc
Peanut Butter Cubes
Eclairs
99% Dark & Pure
85% Dark Bars
Organic Choco Syrup
Spicy Special Slims
Mint Chip Choco
Almond Choco
Manuka Honey Choco
Milk Bars
Raspberry Choco
After Nines
Fruit & Nut Bars
Drinking Coco
Orange Choco
Baker's Choco Chips
Choco Coated Almonds
Caramel Stuffed Bars
70% Dark Bites
Product

1.3 3. Data Preprocessing

[297]:
```
daily_sales = df.groupby('Date')['Amount'].sum().reset_index()
daily_sales = daily_sales.sort_values('Date')
```

[302]:
```
daily_sales.columns = ["ds", "y"]
```

1.4 4. Model Evaluation

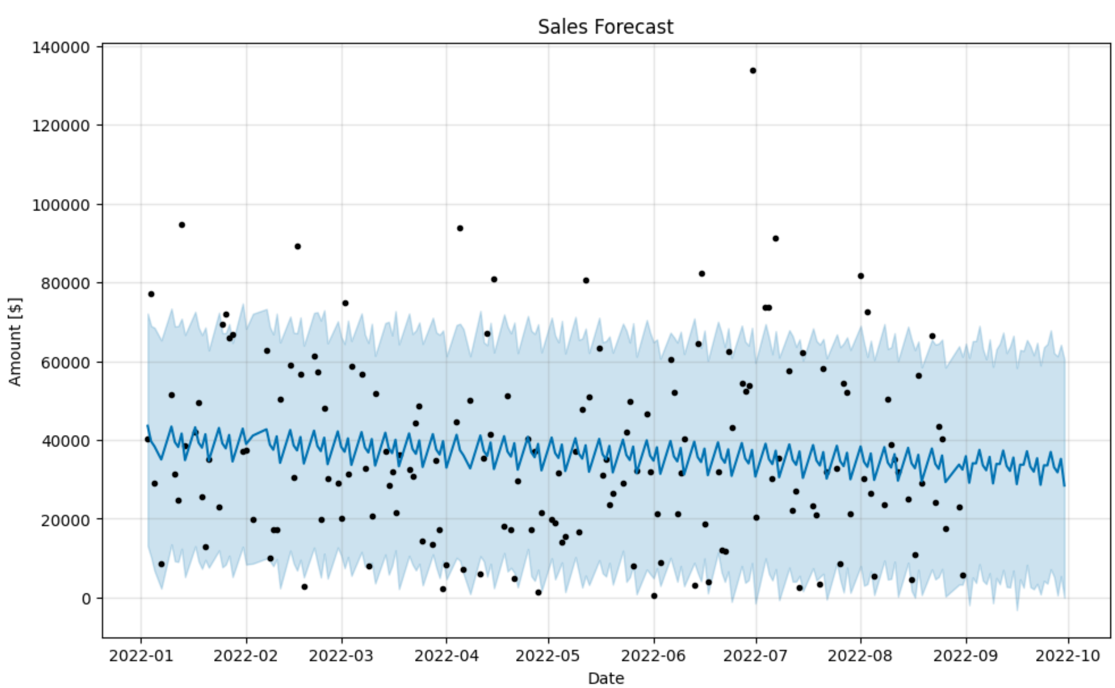
[303]:
```
model_prophet  = Prophet()
model_prophet.fit(daily_sales)
```

```
14:55:37 - cmdstanpy - INFO - Chain [1] start processing
14:55:37 - cmdstanpy - INFO - Chain [1] done processing
```

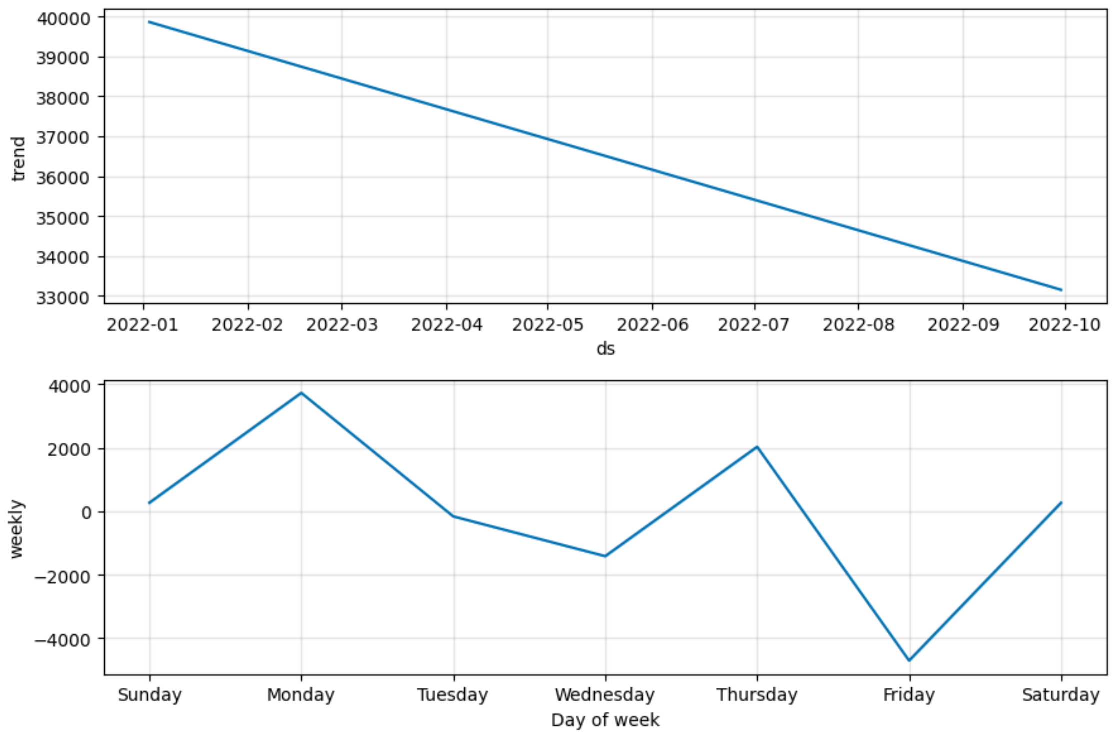
[303]: <prophet.forecaster.Prophet at 0x266c21a8bf0>

[304]:
```
future = model_prophet.make_future_dataframe(periods=30)
forecast = model_prophet.predict(future)
```

[305]:
```
fig1 = model_prophet.plot(forecast)
plt.title("Sales Forecast")
plt.xlabel("Date")
plt.ylabel("Amount [$]")
plt.show()
```



[306]:
```
fig2 = model_prophet.plot_components(forecast)
plt.show()
```



[307]:
```
df_cv = cross_validation(model_prophet, initial='180 days', period='60 days',␣
  ↪horizon='30 days')
df_metrics = performance_metrics(df_cv)
df_metrics.head()
```

```
  0%|          | 0/1 [00:00<?, ?it/s]

14:55:40 - cmdstanpy - INFO - Chain [1] start processing
14:55:40 - cmdstanpy - INFO - Chain [1] done processing
```

[307]:
```
  horizon           mse          rmse           mae      mape     mdape  \
0  2 days  6.898719e+08  26265.412679  22679.382862  0.404227  0.404227
1  3 days  7.346350e+08  27104.151742  24643.504853  0.500415  0.500415
2  4 days  5.201394e+08  22806.564626  21357.939156  3.008227  3.008227
3  7 days  6.012114e+08  24519.611630  23906.162264  3.145816  3.145816
4  8 days  2.284358e+08  15114.093414  14618.992754  0.497319  0.497319

      smape  coverage
```

```
0  0.464535       0.5
1  0.530920       0.5
2  0.935494       0.5
3  1.014486       0.5
4  0.400526       1.0
```

2 5. Conclusions

The chocolate sales dataset does not show clear seasonality, although a decreasing trend over time is observed. Prophet forecasting also suggests that sales may continue to decline. However, the dataset presents high variability, making the series relatively difficult to predict accurately.

[ ]:
```
```